Ratios, rates, proportional relationships, and units answer key

#	ID	Answer	Difficulty
1	9594986d-588a-4127-87f9-41a6f360b9e1	C	Easy
2	ae2818a7-d027-48ec-a6f6-d9167cb25167	348	Easy
3	10654-DC	A	Easy
4	d6f9ca5e-947e-4a5c-a0d0-de08d17d11f5	2520	Easy
5	029578-DC	D	Easy
6	029322-DC	?	Easy
7	c11c7138-6922-4b01-8ac3-981d24884e18	B	Easy
8	04108-DC	A	Easy
9	a9c40475-9ebc-41fc-bfa0-50236b060e50	A	Easy
10	bc2db446-6593-43b2-b2b0-2932e35e1153	A	Easy
11	05121-DC	C	Easy
12	47ff0dd5-480f-4331-b9a9-ee44e9b40d9e	B	Easy
13	022849-DC	?	Easy
14	d88daa3d-713d-4905-8fc3-540215e519e9	A	Easy
15	a1a5d2fd-4a30-4105-b4de-ee544853263c	A	Easy
16	973f54f3-20e6-4be8-a0e8-f04182a1dc93	423.5, 847/2	Easy
17	9ff69db4-88d5-4c3b-ad95-8bd4d0c3f462	A	Easy
18	031618-DC	A	Easy
19	b5f08f84-0852-405a-b44a-c1ad327e11dd	66	Easy
20	f715f428-619d-426d-993d-afd4cdeeb5d5	C	Easy

21	029014-DC	D	Easy
22	70312974-5b98-4396-b51e-4f590919f407	A	Easy
23	69320716-58db-43ad-b02d-1664b3dfb359	C	Easy
24	5979deaa-edcc-42e7-8d12-24a01cec2c7a	9	Easy
25	9d4083e9-df43-4c60-9f48-cb64e2d3dc9d	14.1	Easy
26	f9b50fe8-25d0-4faa-ad54-7ba7d01af7db	C	Easy
27	334db357-0b99-4f13-a766-9dde9a462af3	4	Easy
28	025315-DC	B	Easy
29	ee5f324c-581b-4818-b0a7-48d75fb59f22	B	Easy
30	872cefa5-14e6-4009-b839-96c77f5f3579	D	Easy
31	c40a706f-ca84-432c-bdb2-58a4920cbaab	A	Easy
32	a824dc18-95e2-4efa-b635-884db6f2eeef	A	Easy
33	d6f8cc02-96fe-42b9-bfb6-5bd07411ccfc	D	Easy
34	80f12ef5-a97c-4f43-a382-5d490d55eb90	39000	Easy
35	313142de-ba5a-45df-9ffe-0993fba11dce	102	Easy
36	4c11262b-41b4-4c46-84a0-c2f957e8f9e5	31	Easy
37	688b466f-c743-40b5-8bd7-7b8695af38e7	A	Easy
38	444c8744-ce2f-4537-88b4-072fc2d44492	C	Easy
39	1aadad19-c839-4ddb-aa09-87de2e47212a	A	Easy
40	028058-DC	A	Easy
41	c99660ad-c63a-459b-a649-55773967d912	C	Easy
42	eae9c7b8-873d-4ff8-92bd-dffd8ccec2b8	180	Easy
43	7904b1f4-ccaf-441c-ae58-d8dc651e91cc	C	Medium
44	3d00b62c-b08f-45da-ac4b-7750ae794184	A	Medium
45	72a6fbc6-5106-42d1-b891-58079546b368	D	Medium
46	14884-DC	D	Medium

47	070608-DC	?	Medium
48	c53c49b5-ed37-4cd3-b08c-7994919f7e09	90	Medium
49	252ab91d-897c-4c0d-8d50-e60456359300	B	Medium
50	27dab70d-b227-496d-9793-7bbb537c6638	17	Medium
51	e2e5fe90-eac5-48a3-a466-f4a6ead5776e	73920	Medium
52	98ff2d18-fd0b-4125-ae68-dffac04ee1ef	567	Medium
53	d2f59729-f3aa-4681-adb0-ddf2309976b8	35728	Medium
54	65c97795-cef4-4246-9fe9-82f67f8f7823	16606	Medium
55	028848-DC	A	Medium
56	021174-DC	A	Medium
57	bc6c4c12-4c9a-4382-963e-957612ede87c	D	Medium
58	e098ab1f-ca25-4df4-862e-fe8448392a67	A	Medium
59	e7ba7b4b-ddc3-498e-9508-402688a85aef	980	Medium
60	01f43331-d478-491e-90be-1bf336840ec3	370	Medium
61	070902-DC	B	Medium
62	d4996faa-e699-48e5-8816-df8bd4c0420c	2432	Medium
63	295c177c-ee26-49e1-8364-8c9b6677b31a	24	Medium
64	029306-DC	A	Medium
65	10619-DC	B	Medium
66	021593-DC	D	Medium
67	cefd9f5a-cd0c-4e0f-b298-90f990380e14	B	Medium
68	7d9689ed-c13d-4213-a6f7-685ae6b402ee	34672	Medium
69	019681-DC	D	Medium
70	b32d7197-2b76-4429-8221-4e5482d511ae	1015	Medium
71	46ab20b9-f632-4d8f-a085-104d8f446b46	14/3, 4.666, 4.667	Medium
72	e2565394-6b89-4d8e-8b02-06a363d8a20f	40260	Medium

73	028911-DC	A	Medium
74	070624-DC	?	Medium
75	16d02cf9-fae4-4096-a277-a897cfed31be	A	Hard
76	93a3d90a-8e47-405b-bf18-8a2d7da11a63	D	Hard
77	3a321846-3dc0-4a6d-82e6-2cfcf81ef623	A	Hard
78	933baf7b-cfc3-42f9-a9f8-f184d6f97c27	B	Hard
79	d6f5d386-dd70-4e85-96cc-2c0859c39acc	A	Hard
80	025175-DC	D	Hard
81	ea65c30d-d2d3-4c66-b79e-881cd28208cf	D	Hard
82	964d5413-e030-4000-a612-54941886f548	B	Hard
83	c084b0a0-2744-443d-9e6a-f230e22fdc88	0.32, 8/25	Hard
84	cd36e81a-3f06-40da-ab13-b7cd41cd0e9b	A	Hard
85	03862-DC	?	Hard
86	86610db9-1732-40a7-a795-6d71991b27f8	6.21	Hard
87	837ca3a4-1a14-44b1-84b5-5f2a706e30f9	.0625, 1/16	Hard
88	ec4a70f3-71cf-4dca-9776-c588830c5d34	C	Hard
89	a9ed5978-3ec1-4d21-966b-82a5c7c417f3	4176	Hard
90	8452380b-8279-44a6-98f5-7b0beb6adc6e	D	Hard
91	74b99987-9634-407d-adbe-a305f02a1c5f	157.8, 789/5	Hard
92	56513b01-fe94-443b-8898-f5643702eab1	B	Hard
93	fc00704d-6b53-445f-a379-309a3e911067	B	Hard

Rationales

▼ 1 — C

Choice C is correct. Since 1 meter is equal to 100 centimeters, 51 meters is equal to $51 \text{ meters} \frac{100 \text{ centimeters}}{1 \text{ meter}}$, or $5{,}100$ centimeters.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from dividing, rather than multiplying, 51 by 100.

Choice D is incorrect. This is the length, in millimeters rather than centimeters, that is equivalent to a length of 51 meters.

▼ 2 — 348

The correct answer is 348. It's given that 1 fathom is equivalent to 6 feet. Therefore, 58 fathoms is equivalent to $58 \text{ fathoms} \frac{6 \text{ feet}}{1 \text{ fathom}}$, or 348 feet. Thus, when the camera is at a depth of 58 fathoms, the camera's depth, in feet, is 348.

▼ 3 — A

Choice A is correct. At the exchange rate of 1 peso = 0.075 US dollars, 7,500 pesos would be converted to 7,500 × 0.075 = $562.50. However, since Maria pays a 2% fee on the converted US dollar amount, she receives only (100 – 2)%, or 98%, of the converted US dollars, and 562.50 × 0.98 = $551.25.

Choice B is incorrect. This is the number of US dollars Maria would receive if the exchange service did not charge a 2% fee. Choice C is incorrect and may result from a decimal point error made when calculating the conversion to US dollars and from not assessing the 2% fee. Choice D is incorrect and may result from reversing the units of the exchange rate.

▼ 4 — 2520

The correct answer is $2{,}520$. There are 60 minutes in one hour. At a rate of 42 posters per minute, the number of posters produced in one hour can be determined by $\frac{42 \text{ posters}}{1 \text{ minute}} \frac{60 \text{ minutes}}{1 \text{ hour}}$, which is $2{,}520$ posters per hour.

▼ 5 — D

Choice D is correct. Since the ratio of y to x is constant for each ordered pair in the table, the first row can be used to determine that the ratio of y to x is 4 to 1. The proportion $\frac{4}{1} = \frac{k}{40}$ can be used to solve for k. Multiplying each side of the equation by 40 yields $160 = k$.

Choice A is incorrect. This is the value of y when the value of x is 7, not 40. Choice B is incorrect and may result from subtracting 4 from 40 instead of multiplying 40 by 4. Choice C

is incorrect and may result from incorrectly setting up the proportion.

▼ 6 — ?

Choice D is correct. It's given that the ratio of m to k is estimated to be 1 to 5. Therefore, when $k = 25$, the relationship between these ratios can be expressed by the proportion $\frac{m}{25} = \frac{1}{5}$. Multiplying both sides of this equation by 25 yields $m = 5$.

Choices A, B, and C are incorrect and may result from calculation errors.

▼ 7 — B

Choice B is correct. It's given that 100 centimeters is equal to 1 meter. Therefore, 2,300 centimeters is equivalent to $2{,}300 \text{ centimeters} \frac{1 \text{ meter}}{100 \text{ centimeters}}$, or 23 meters.

Choice A is incorrect. 0.043 meters is equivalent to 4.3, not 2,300, centimeters.

Choice C is incorrect. 2,400 meters is equivalent to 240,000, not 2,300, centimeters.

Choice D is incorrect. 230,000 meters is equivalent to 23,000,000, not 2,300, centimeters.

▼ 8 — A

Choice A is correct. If 2.5 ounces of chocolate are needed for each muffin, then the number of ounces of chocolate needed to make 48 muffins is $48 \times 2.5 = 120$ ounces. Since 1 pound = 16 ounces, the number of pounds that is equivalent to 120 ounces is $\frac{120}{16} = 7.5$ pounds. Therefore, 7.5 pounds of chocolate are needed to make the 48 muffins.

Choice B is incorrect. If 10 pounds of chocolate were needed to make 48 muffins, then the total number of ounces of chocolate needed would be $10 \times 16 = 160$ ounces. The number of ounces of chocolate per muffin would then be $\frac{160}{48} = 3.33$ ounces per muffin, not 2.5 ounces per muffin. Choices C and D are also incorrect. Following the same procedures as used to test choice B gives 16.8 ounces per muffin for choice C and 40 ounces per muffin for choice D, not 2.5 ounces per muffin. Therefore, 50.5 and 120 pounds cannot be the number of pounds needed to make 48 signature chocolate muffins.

▼ 9 — A

Choice A is correct. If the object travels 108 centimeters at a speed of 12 centimeters per second, the time of travel can be determined by dividing the total distance by the speed. This results in $\frac{108 \text{ centimeters}}{12 \text{ centimeters/second}}$, which is 9 seconds.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 10 — A

Choice A is correct. It's given that a special camera is used for underwater ocean research, and this camera is at a depth of 39 fathoms. It's also given that 1 fathom is equal to 6 feet. Thus, 39 fathoms is equivalent to $39 \text{ fathoms} \frac{6 \text{ feet}}{1 \text{ fathom}}$, or 234 feet. Therefore, the camera's depth, in feet, is 234.

Choice B is incorrect. This is the camera's depth, in feet, if the camera is at a depth of 19.5 fathoms.

Choice C is incorrect. This is the camera's depth, in feet, if the camera is at a depth of 7.5 fathoms.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 11 — C

Choice C is correct. The weight of an object on Venus is approximately $\frac{9}{10}$ of its weight on Earth. If an object weighs 100 pounds on Earth, then the object's weight on Venus is approximately $\frac{9}{10}(100) = 90$ pounds. The same object's weight on Jupiter is approximately $\frac{23}{10}$ of its weight on Earth; therefore, the object weighs approximately $\frac{23}{10}(100) = 230$ pounds on Jupiter. The difference between the object's weight on Jupiter and the object's weight on Venus is approximately $230 - 90 = 140$ pounds. Therefore, an object that weighs 100 pounds on Earth weighs 140 more pounds on Jupiter than it weighs on Venus.

Choice A is incorrect because it is the weight, in pounds, of the object on Venus. Choice B is incorrect because it is the weight, in pounds, of an object on Earth if it weighs 100 pounds on Venus. Choice D is incorrect because it is the weight, in pounds, of the object on Jupiter.

▼ 12 — B

Choice B is correct. It's given that the fish swam $5{,}104$ yards and that 1 mile is equal to $1{,}760$ yards. Therefore, the fish swam $5{,}104 \text{ yards} \frac{1 \text{ mile}}{1{,}760 \text{ yards}}$, which is equivalent to $\frac{5{,}104}{1{,}760}$ miles, or 2.9 miles.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 13 — ?

Choice C is correct. If the butterfly traveled 2,100 miles in 120 days, then it traveled, on average, $\frac{2{,}100 \text{ miles}}{120 \text{ days}} = 17.5$ miles per day.

Choice A is incorrect. This is approximately the average amount of time, in days, it took the butterfly to fly one mile: $\frac{120 \text{ days}}{2{,}100 \text{ miles}} = 0.057$ days per mile. Choice B is incorrect and may result from an arithmetic error. Choice D is incorrect. This is the number of hours in a day rather than the number of miles flown per day.

▼ 14 — A

Choice A is correct. It's given that the ratio of black pens to red pens is 8 to 1. Therefore, there are $\frac{1}{8}$ as many red pens as black pens in the box. It's also given that there are 40 black pens in the box. Therefore, the number of red pens is $\frac{1}{8}$ of the 40 black pens. Thus, the number of red pens is $40\frac{1}{8}$, or 5.

Choice B is incorrect. This is the number of black pens in the box for every red pen.

Choice C is incorrect. This is the number of black pens in the box.

Choice D is incorrect. This is the number of red pens in the box if the ratio of black pens to red pens is 1 to 8.

▼ 15 — A

Choice A is correct. It's given that Tilly earns p dollars for every w hours of work. This can be represented by the proportion $\frac{p}{w}$. The amount of money, x, Tilly earns for $39w$ hours of work

can be found by setting up the proportion $\frac{p}{w} = \frac{x}{39w}$. This can be rewritten as $39pw = xw$. Dividing both sides by w results in $x = 39p$.

Choice B is incorrect. This is the amount of money Tilly earns in dollars per hour, not the amount of money Tilly earns for $39w$ hours of work.

Choice C is incorrect. This is the amount of money Tilly earns for w hours of work plus 39, not the amount of money Tilly earns for $39w$ hours of work.

Choice D is incorrect. This is the amount of money Tilly earns for w hours of work minus 39, not the amount of money Tilly earns for $39w$ hours of work.

▼ 16 — 423.5, 847/2

The correct answer is 423.5. It's given that $5.5 \text{ yards} = 1 \text{ rod}$. Therefore, 77 rods is equivalent to $77 \text{ rods} \frac{5.5 \text{ yards}}{1 \text{ rod}}$, or 423.5 yards. Note that 423.5 and 847/2 are examples of ways to enter a correct answer.

▼ 17 — A

Choice A is correct. It's given that the ratio x to y is equivalent to the ratio 12 to t. This can be represented by $\frac{x}{y} = \frac{12}{t}$. Substituting 156 for x in this equation yields $\frac{156}{y} = \frac{12}{t}$. This can be rewritten as $12y = 156t$. Dividing both sides of this equation by 12 yields $y = 13t$. Therefore, when $x = 156$, the value of y in terms of t is $13t$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 18 — A

Choice A is correct. It's given that $t = 4u$. Multiplying both sides of this equation by 2 yields $2t = 2(4u)$, or $2t = 8u$.

Choice B is incorrect and may result from dividing, instead of multiplying, the right-hand side of the equation by 2. Choices C and D are incorrect and may result from calculation errors.

▼ 19 — 66

The correct answer is 66. It's given that a product costs 11.00 dollars per pound. Therefore, the cost for 6 pounds of the product is $\frac{11.00 \text{ dollars}}{1 \text{ pound}} 6 \text{ pounds}$, which is equivalent to 66.00, or 66, dollars.

▼ 20 — C

Choice C is correct. It's given that 3 teaspoons is equivalent to 1 tablespoon. Therefore, 44 tablespoons is equivalent to $44 \text{ tablespoons} \frac{3 \text{ teaspoons}}{1 \text{ tablespoon}}$, or 132 teaspoons.

Choice A is incorrect. This is equivalent to approximately 15.66 tablespoons, not 44 tablespoons.

Choice B is incorrect. This is equivalent to approximately 29.33 tablespoons, not 44 tablespoons.

Choice D is incorrect. This is equivalent to approximately 58.66 tablespoons, not 44 tablespoons.

▼ 21 — D

Choice D is correct. It's given that the ratio of t to u is 1 to 2. Since $t = 10$, it follows that the ratio of 10 to u is also 1 to 2. The relationship between these ratios can be represented by the proportion $\frac{10}{u} = \frac{1}{2}$. Multiplying both sides of this equation by 2 and then by u yields $20 = u$.

Choice A is incorrect. This is the value of u when $t = 1$. Choice B is incorrect. This would be the value of u if the ratio of t to u were 2 to 1. Choice C is incorrect. This is the value of t, not u.

▼ 22 — A

Choice A is correct. It's given that a certain bird species can fly at an average speed of 16 meters per second when in continuous flight. At this rate, in 4 seconds this bird species would fly $\frac{16 \text{ meters}}{\text{second}} 4 \text{ seconds}$, or 64 meters.

Choice B is incorrect. This is the value of 16 + 4, not 164.

Choice C is incorrect. This is the distance the bird would fly in 1 second, not 4 seconds.

Choice D is incorrect. This is the value of 16 - 4, not 164.

▼ 23 — C

Choice C is correct. It's given that the population density of Worthington is 290 people per square mile and Worthington has a population of $92{,}800$ people. Therefore, the area of Worthington is $92{,}800 \text{ people} \frac{1 \text{ square mile}}{290 \text{ people}}$, which is equivalent to $\frac{92{,}800 \text{ square miles}}{290}$, or 320 square miles.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 24 — 9

The correct answer is 9. It's given that the customer spent $\$ 27$ to purchase oranges at $\$ 3$ per pound. Therefore, the number of pounds of oranges the customer purchased is $\$ 27 \frac{1 \text{ pound}}{\$ 3}$, or 9 pounds.

▼ 25 — 14.1

The correct answer is 14.1. It's given that a participant completes the bicycle race with an average speed of $24{,}816$ yards per hour and $1 \text{ mile} = 1{,}760 \text{ yards}$. It follows that this average speed is equivalent to $\frac{24{,}816 \text{ yards}}{1 \text{ hour}} \frac{1 \text{ mile}}{1{,}760 \text{ yards}}$, which yields $\frac{14.1 \text{ miles}}{1 \text{ hour}}$, or 14.1 miles per hour.

▼ 26 — C

Choice C is correct. It's given that 8 out of 300 customers in a random sample said they would be interested in the new plan. This means the fraction of sampled customers who said they would be interested is $\frac{8}{300}$. Multiplying this fraction by the total customer population yields $(\frac{8}{300})(30{,}000)$, or 800. Therefore, the best estimate of the total number of customers who would be interested in the new service plan is 800.

Choice A is incorrect. This is the number of customers surveyed who said they would be interested in the new plan, not the best estimate of the total number of customers who would be interested in the new plan.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 27 — 4

The correct answer is 4. It's given that the object travels at a constant speed of 6 centimeters per second. The speed of the object can be written as $\frac{6 \text{ centimeters}}{1 \text{ second}}$. Let x represent the time, in seconds, it would take for the object to travel 24 centimeters. The value of x can be calculated by solving the equation $\frac{6 \text{ centimeters}}{1 \text{ second}} = \frac{24 \text{ centimeters}}{x \text{ seconds}}$, which can be written as $\frac{6}{1} = \frac{24}{x}$, or $6 = \frac{24}{x}$. Multiplying each side of this equation by x yields $6x = 24$. Dividing each side of this equation by 6 yields $x = 4$. Therefore, it would take the object 4 seconds to travel 24 centimeters.

▼ 28 — B

Choice B is correct. It's given that the event will be in a 5,400-square-foot room and that there should be at least 8 square feet per person. The maximum number of people that could attend the event can be found by dividing the total square feet in the room by the minimum number of square feet needed per person, which gives $\frac{5,400}{8} = 675$.

Choices A and C are incorrect and may result from conceptual or computational errors. Choice D is incorrect and may result from multiplying, rather than dividing, 5,400 by 8.

▼ 29 — B

Choice B is correct. It's given that $1 \text{ yard} = 36 \text{ inches}$. Therefore, 612 inches is equivalent to $612 \text{ inches} \frac{1 \text{ yard}}{36 \text{ inches}}$, which can be rewritten as $\frac{612 \text{ yards}}{36}$, or 17 yards.

Choice A is incorrect. This is the number of yards that are equivalent to 2.124 inches.

Choice C is incorrect. This is the number of yards that are equivalent to 20,736 inches.

Choice D is incorrect. This is the number of yards that are equivalent to 793,152 inches.

▼ 30 — D

Choice D is correct. It's given that 4 cups is equal to 1 quart. Multiplying $\frac{4 \text{ cups}}{1 \text{ quart}}$ by 83 quarts yields $4 \cdot 83$ cups, or 332 cups.

Choice A is incorrect. This is the number of quarts in 83 cups, rounded to the nearest whole number.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 31 — A

Choice A is correct. It's given that a kangaroo has a mass of 28 kilograms and that 1 kilogram is equal to $1{,}000$ grams. Therefore, the kangaroo's mass, in grams, is $28 \text{ kilograms} \frac{1{,}000 \text{ grams}}{1 \text{ kilogram}}$, which is equivalent to $28{,}000$ grams.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 32 — A

Choice A is correct. Dividing the number of revolutions by the number of minutes gives the number of revolutions the turbine completes per minute. It's given that the wind turbine completes 900 revolutions in 50 minutes. Therefore, at this rate, this turbine completes $\frac{900}{50}$, or 18, revolutions per minute.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 33 — D

Choice D is correct. Since 1 yard is equal to 3 feet, 64 yards is equal to $64 \text{ yards} \frac{3 \text{ feet}}{1 \text{ yard}}$, or 192 feet. It follows that 64 yards per second is equivalent to 192 feet per second. Therefore, the object's speed is 192 feet per second.

Choice A is incorrect. A speed of 61 feet per second is equivalent to $\frac{61}{3}$, not 64, yards per second.

Choice B is incorrect. A speed of 67 feet per second is equivalent to $\frac{67}{3}$, not 64, yards per second.

Choice C is incorrect. A speed of 94 feet per second is equivalent to $\frac{94}{3}$, not 64, yards per second.

▼ 34 — 39000

The correct answer is $39{,}000$. It's given that the giant armadillo has a mass of 39 kilograms. Since 1 kilogram is equal to $1{,}000$ grams, 39 kilograms is equal to $39 \text{ kilograms} \frac{1{,}000 \text{ grams}}{1 \text{ kilogram}}$, or $39{,}000$ grams. Therefore, the giant armadillo's mass, in grams, is $39{,}000$.

▼ 35 — 102

The correct answer is 102. It's given that 1 yard is equivalent to 3 feet. Therefore, 34 yards is equivalent to $34 \text{ yards} \frac{3 \text{ feet}}{1 \text{ yard}}$, or 102 feet.

▼ 36 — 31

The correct answer is 31. It's given that 1 yard is equal to 36 inches. Therefore, $1{,}116$ inches is equivalent to $1{,}116 \text{ inches} \frac{1 \text{ yard}}{36 \text{ inches}}$, or 31 yards.

▼ 37 — A

Choice A is correct. It's given that the ratio of the length of line segment XY to the length of line segment ZV is 6 to 1, which means $\frac{XY}{ZV} = \frac{6}{1}$. It's given that the length of line segment XY is 102 inches. If the length, in inches, of line segment ZV is represented by l, the value of l can be calculated by solving the equation $\frac{102}{l} = \frac{6}{1}$, or $\frac{102}{l} = 6$. Multiplying each side of this equation by l yields $102 = 6l$. Dividing each side of this equation by 6 yields $17 = l$. Therefore, the length of line segment ZV is 17 inches.

Choice B is incorrect. This is the length, in inches, of line segment ZV if the length of line segment XY is 576, not 102, inches.

Choice C is incorrect. This is the length, in inches, of line segment XY, not line segment ZV.

Choice D is incorrect. This is the length, in inches, of line segment ZV if the ratio of the length of line segment XY to the length of line segment ZV is 1 to 6, not 6 to 1.

▼ 38 — C

Choice C is correct. It's given that $1 \text{ centimeter} = 10 \text{ millimeters}$. It follows that a velocity of 76 centimeters per second is equivalent to $\frac{76 \text{ centimeters}}{1 \text{ second}} \frac{10 \text{ millimeters}}{1 \text{ centimeter}}$, which is equivalent to 7610, or 760, millimeters per second.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 39 — A

Choice A is correct. It's given that there are $2{,}358$ raccoons in a 131-square-mile area. The estimated population density, in raccoons per square mile, is the estimated number of raccoons divided by the number of square miles. Therefore, the estimated population density of this area is $\frac{2{,}358 \text{ raccoons}}{131 \text{ square miles}}$, or 18 raccoons per square mile.

Choice B is incorrect. This is the number of square miles in the area, not the estimated number of raccoons per square mile in this area.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 40 — A

Choice A is correct. It's given that Shaquan has 7 red cards and 28 blue cards. Therefore, the ratio of red cards to blue cards that Shaquan has is 7 to 28. This ratio can be reduced by dividing both parts of the ratio by 7, which yields the ratio 1 to 4.

Choice B is incorrect. This is the ratio of blue cards to red cards that Shaquan has. Choice C is incorrect and may result from a calculation error when reducing the ratio. Choice D is incorrect. This may result from finding the ratio of blue cards to red cards, or 28 to 7, and then making a calculation error when reducing the ratio.

▼ 41 — C

Choice C is correct. It's given that the cherry pitting machine pits 12 pounds of cherries in 3 minutes. This rate can be written as $\frac{12 \text{ pounds of cherries}}{3 \text{ minutes}}$. If the number of minutes it takes the machine to pit 96 pounds of cherries is represented by x, the value of x can be calculated by solving the equation $\frac{12 \text{ pounds of cherries}}{3 \text{ minutes}} = \frac{96 \text{ pounds of cherries}}{x \text{ minutes}}$, which can be rewritten as $\frac{12}{3} = \frac{96}{x}$, or $4 = \frac{96}{x}$. Multiplying each side of this equation by x yields $4x = 96$. Dividing each side of this equation by 4 yields $x = 24$. Therefore, it takes the machine 24 minutes to pit 96 pounds of cherries.

Choice A is incorrect. This is the number of minutes it takes the machine to pit 32, not 96, pounds of cherries.

Choice B is incorrect. This is the number of minutes it takes the machine to pit 60, not 96, pounds of cherries.

Choice D is incorrect. This is the number of minutes it takes the machine to pit 144, not 96, pounds of cherries.

▼ 42 — 180

The correct answer is 180. It's given that a mechanical device produces items at a constant rate of 60 items per hour. This rate can be written as $\frac{60 \text{ items}}{1 \text{ hour}}$. Let x represent the number of items the mechanical device will produce in 3 hours at the given rate. It follows that $\frac{60 \text{ items}}{1 \text{ hour}} = \frac{x \text{ items}}{3 \text{ hours}}$, which can be written as $\frac{60}{1} = \frac{x}{3}$, or $60 = \frac{x}{3}$. Multiplying each side of this equation by 3 yields $180 = x$. Therefore, at the given rate, the mechanical device will produce 180 items in 3 hours.

Alternate approach: It's given that a mechanical device produces items at a constant rate of 60 items per hour. At this rate, the mechanical device will produce $\frac{60 \text{ items}}{1 \text{ hour}} 3 \text{ hours}$, or 180 items in 3 hours.

▼ 43 — C

Choice C is correct. It's given that there is a proportional relationship between p and t. This relationship can be written as $\frac{p}{t}$. If $p = 5{,}130$ when $t = 6{,}840$, the value of p when $t = 5{,}472$ can be calculated by solving the equation $\frac{5{,}130}{6{,}840} = \frac{p}{5{,}472}$. Multiplying each side of this equation by $5{,}472$ yields $5{,}472\frac{5{,}130}{6{,}840} = \frac{p}{5{,}472}5{,}472$, or $p = 4{,}104$. Therefore, when $t = 5{,}472$, the value of p is $4{,}104$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 44 — A

Choice A is correct. It's given that the object has a mass of 168 grams and a volume of 24 cubic centimeters. Dividing the mass, in grams, of the object by the volume, in cubic centimeters, of the object gives the density, in grams per cubic centimeter, of the object. It follows that the density of the object is $\frac{168 \text{ grams}}{24 \text{ cubic centimeters}}$, which is equivalent to $\frac{168}{24}$ grams per cubic centimeter, or 7 grams per cubic centimeter.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 45 — D

Choice D is correct. It's given that the total mass, in kilograms, of r identical objects is t. To obtain the expression $146r$, r is multiplied by 146. Therefore, to find the total mass, in kilograms, of $146r$ of these objects, t must also be multiplied by 146. The result of multiplying t by 146 is the expression $146t$. Therefore, the total mass, in kilograms, of $146r$ of these objects, is $146t$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

▼ 46 — D

Choice D is correct. The cost per gram of protein in 1 large egg is \$0.36 ÷ 6 = \$0.06. The cost per gram of protein in 1 cup of milk is \$0.24 ÷ 8 = \$0.03. It follows that the ratio of the cost per gram of protein in a large egg to the cost per gram of protein in a cup of milk is 0.06:0.03, which can be rewritten as 2:1.

Choice A is incorrect and may result from finding the ratio of the cost per gram of protein in a cup of milk to the cost per gram of protein in a large egg (the reciprocal of the ratio specified in the question). Choices B and C are incorrect and may result from incorrectly calculating the unit rates or from errors made when simplifying the ratio.

▼ 47 — ?

The correct answer is 4.44. The selling price, in dollars per cubic inch, is found by multiplying the density, in ounces per cubic inch, by the unit price, in dollars per ounce: $\left(\frac{0.555 \text{ ounce}}{1 \text{ cubic inch}}\right)\left(\frac{\$8.00}{1 \text{ ounce}}\right)$ yields $\frac{\$4.44}{1 \text{ cubic inch}}$. Thus, the selling price, in dollars per cubic inch, is 4.44.

▼ 48 — 90

The correct answer is 90. It's given that the ratio of x to y is equivalent to the ratio 9 to 5. It follows that $\frac{x}{y} = \frac{9}{5}$. Multiplying each side of this equation by $5y$ yields $\frac{5yx}{y} = \frac{95y}{5}$, or $5x = 9y$. Dividing each side of this equation by 9 yields $\frac{5x}{9} = y$. Substituting 162 for x in this equation yields $\frac{5162}{9} = y$, which is equivalent to $\frac{810}{9} = y$, or $90 = y$. Therefore, if the value of x is 162, the value of y is 90.

▼ 49 — B

Choice B is correct. It's given that the ratio of the rectangular region's length to its width is 35 to 10. This can be written as a proportion: $\frac{\text{length}}{\text{width}} = \frac{35}{10}$, or $\frac{l}{w} = \frac{35}{10}$. This proportion can be rewritten as $10l = 35w$, or $l = 3.5w$. If the width of the rectangular region increases by 7, then the length will increase by some number x in order to maintain this ratio. The value of x can be found by replacing l with $l + x$ and w with $w + 7$ in the equation, which gives $l + x = 3.5w + 7$. This equation can be rewritten using the distributive property as $l + x = 3.5w + 24.5$. Since $l = 3.5w$, the right-hand side of this equation can be rewritten by substituting l for $3.5w$, which gives $l + x = l + 24.5$, or $x = 24.5$. Therefore, if the width of the rectangular region increases by 7 units, the length must increase by 24.5 units in order to maintain the given ratio.

Choice A is incorrect. If the width of the rectangular region increases, the length must also increase, not decrease.

Choice C is incorrect. If the width of the rectangular region increases, the length must also increase, not decrease.

Choice D is incorrect. Since the ratio of the length to the width of the rectangular region is 35 to 10, if the width of the rectangular region increases by 7 units, the length would have to increase by a proportional amount, which would have to be greater than 7 units.

▼ 50 — 17

The correct answer is 17. If one value is multiplied by a number, then the other value must be multiplied by the same number in order to maintain the same ratio. It's given that j is multiplied by 17. Therefore, in order to maintain the same ratio, k must also be multiplied by 17.

▼ 51 — 73920

The correct answer is 73,920. It's given that $1 \text{ furlong} = 220 \text{ yards}$ and $1 \text{ yard} = 3 \text{ feet}$. It follows that a distance of 112 furlongs is equivalent to $112 \text{ furlongs} \frac{220 \text{ yards}}{1 \text{ furlong}} \frac{3 \text{ feet}}{1 \text{ yard}}$, or 73,920 feet.

▼ 52 — 567

The correct answer is 567. It's given that the ratio a to b is equivalent to the ratio 63 to 17, or $\frac{a}{b} = \frac{63}{17}$. When $b = 153$, the value of a can be calculated by substituting 153 for b in the equation $\frac{a}{b} = \frac{63}{17}$, which yields $\frac{a}{153} = \frac{63}{17}$. Multiplying both sides of this equation by 153 yields $a = 567$. Thus, when $b = 153$, the value of a is 567.

▼ 53 — 35728

The correct answer is 35,728. It's given that $1 \text{ mile} = 1{,}760 \text{ yards}$. It follows that an average speed of 20.300 miles per hour is equivalent to $\frac{20.300 \text{ miles}}{1 \text{ hour}} \frac{1{,}760 \text{ yards}}{1 \text{ mile}}$, or 35,728 yards per hour.

▼ 54 — 16606

The correct answer is 16,606. It's given that one side of a flat board has an area of 874 square inches. If a pressure of 19 pounds per square inch of area is exerted on this side of the board, the total force exerted on this side of the board is $874 \text{ square inches} \frac{19 \text{ pounds}}{1 \text{ square inch}}$, or 16,606 pounds.

▼ 55 — A

Choice A is correct. It's given that rectangle A has length 15 and width w. Therefore, the length-to-width ratio of rectangle A is 15 to w. It's also given that rectangle B has length 20 and the same length-to-width ratio as rectangle A. Let x represent the width of rectangle B. The proportion $\frac{15}{w} = \frac{20}{x}$ can be used to solve for x in terms of w. Multiplying both sides of this equation by x yields $\frac{15x}{w} = 20$, and then multiplying both sides of this equation by w yields $15x = 20w$. Dividing both sides of this equation by 15 yields $x = \frac{20w}{15}$. Simplifying this fraction yields $x = \frac{4}{3}w$.

Choices B and D are incorrect and may result from interpreting the difference in the lengths of rectangle A and rectangle B as equivalent to the difference in the widths of rectangle A and rectangle B. Choice C is incorrect and may result from using a length-to-width ratio of w to 15, instead of 15 to w.

▼ 56 — A

Choice A is correct. It's given that when Tanya works z hours, she earns $13.50z$ dollars. Since her hourly rate is constant, if she works 3 times as many hours, or $3z$ hours, she will earn 3 times as many dollars, or $3(13.50z)$.

Choice B is incorrect. This expression represents adding 3 dollars to the $13.50z$ dollars Tanya will earn. Choice C is incorrect. This expression can be rewritten as $16.50z$, which implies that Tanya earns $16.50 per hour, not $13.50. Choice D is incorrect. This expression adds 3 to the number of hours Tanya works, rather than multiplying the hours she works by 3.

▼ 57 — D

Choice D is correct. It's given that $1 \text{ furlong} = 220 \text{ yards}$ and $1 \text{ yard} = 3 \text{ feet}$. It follows that a distance of 354 furlongs is equivalent to $354 \text{ furlongs} \frac{220 \text{ yards}}{1 \text{ furlong}} \frac{3 \text{ feet}}{1 \text{ yard}}$, or 233,640 feet.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 58 — A

Choice A is correct. It's given that $3 \text{ feet} = 1 \text{ yard}$. It follows that a speed of $\frac{3}{20}$ feet per second is equivalent to $\frac{\frac{3}{20} \text{ feet}}{1 \text{ second}} \frac{1 \text{ yard}}{3 \text{ feet}}$, which is equivalent to $\frac{3}{20}\frac{1}{3}$, or $\frac{1}{20}$, yards per second.

Choice B is incorrect. This is the speed, in feet per second, that's equivalent to $\frac{3}{20}$ yards per second.

Choice C is incorrect. This is the speed, in yards per second, that's equivalent to 18, not $\frac{3}{20}$, feet per second.

Choice D is incorrect. This is the speed, in yards per second, that's equivalent to 60, not $\frac{3}{20}$, feet per second.

▼ 59 — 980

The correct answer is 980. It's given that the ratio 140 to m is equivalent to the ratio 4 to 28. Therefore, the value of m can be found by solving the equation $\frac{140}{m} = \frac{4}{28}$. Multiplying each side

of this equation by m yields $140 = \frac{4m}{28}$. Multiplying each side of this equation by 28 yields $3{,}920 = 4m$. Dividing each side of this equation by 4 yields $980 = m$. Therefore, the value of m is 980.

▼ 60 — 370

The correct answer is 370. It's given that the population density of Cedar County is 230 people per square mile and the county has a population of 85,100 people. Based on the population density, it follows that the area of Cedar County is $85{,}100 \text{ people} \frac{1 \text{ square mile}}{230 \text{ people}}$, or 370 square miles.

▼ 61 — B

Choice B is correct. It is given that the volume of the glass is approximately 16 fluid ounces. If Jenny has 1 gallon of water, which is 128 fluid ounces, she could fill the glass $\frac{128}{16} = 8$ times.

Choice A is incorrect because Jenny would need 16×16 fluid ounces = 256 fluid ounces, or 2 gallons, of water to fill the glass 16 times. Choice C is incorrect because Jenny would need only 4×16 fluid ounces = 64 fluid ounces of water to fill the glass 4 times. Choice D is incorrect because Jenny would need only 3×16 fluid ounces = 48 fluid ounces to fill the glass 3 times.

▼ 62 — 2432

The correct answer is 2,432. It's given that $4 \text{ cups} = 1 \text{ quart}$. It follows that 76 quarts is equivalent to $76 \text{ quarts} \frac{4 \text{ cups}}{1 \text{ quart}}$, or 304 cups. It's also given that $8 \text{ fluid ounces} = 1 \text{ cup}$. It follows that 304 cups is equivalent to $304 \text{ cups} \frac{8 \text{ fluid ounces}}{1 \text{ cup}}$, or 2,432 fluid ounces.

▼ 63 — 24

The correct answer is 24. The equation $\frac{24x}{ny} = 4$ can be rewritten as $\frac{24}{n}\frac{x}{y} = 4$. It's given that $\frac{x}{y} = 4$. Substituting 4 for $\frac{x}{y}$ in the equation $\frac{24}{n}\frac{x}{y} = 4$ yields $\frac{24}{n}4 = 4$. Multiplying both sides of this equation by n yields $244 = 4n$. Dividing both sides of this equation by 4 yields $24 = n$. Therefore, the value of n is 24.

▼ 64 — A

Choice A is correct. Since 1 kilometer is equal to 1,000 meters, it follows that 90 kilometers is equal to $90(1{,}000) = 90{,}000$ meters. Since 1 hour is equal to 60 minutes and 1 minute is equal to 60 seconds, it follows that 1 hour is equal to $60(60) = 3{,}600$ seconds. Now $\frac{90 \text{ kilometers}}{1 \text{ hour}}$ is equal to $\frac{90{,}000 \text{ meters}}{3{,}600 \text{ seconds}}$, which reduces to $\frac{25 \text{ meters}}{1 \text{ second}}$ or 25 meters per second.

Choices B, C, and D are incorrect and may result from conceptual or calculation errors.

▼ 65 — B

Choice B is correct. The land area of the coastal city can be found by subtracting the area of the water from the total area of the coastal city; that is, $92.1 - 11.3 = 80.8$ square miles. The population density is the population divided by the land area, or $\frac{621{,}000}{80.8} = 7{,}686$, which is closest to 7,690 people per square mile.

Choice A is incorrect and may be the result of dividing the population by the total area, instead of the land area. Choice C is incorrect and may be the result of dividing the population by the area of water. Choice D is incorrect and may be the result of making a computational error with the decimal place.

▼ 66 — D

Choice D is correct. The increase in Iceland's population can be found by multiplying the increase in population density, in people per square kilometer, by the area, in square kilometers. It's given that the population density of Iceland was 2.5 people per square kilometer in 1990 and 3.3 people per square kilometer in 2014. The increase in population density can be found by subtracting 2.5 from 3.3, which yields 0.8. It's given that the land area of Iceland was 100,250 square kilometers. Thus, the increase in population is $0.8(100{,}250)$, or 80,200.

Alternate approach: It's given that the population density of Iceland, in people per square kilometer of land area, in 1990 was 2.5. Since the land area of Iceland was 100,250 square kilometers, it follows that the population of Iceland in 1990 was $2.5(100{,}250)$, or 250,625. Similarly, the population of Iceland in 2014 was $3.3(100{,}250)$, or 330,825. The population

increase is the difference in the population from 1990 to 2014, or $330{,}825 - 250{,}625$, which yields 80,200. Therefore, Iceland's population increased by 80,200 from 1990 to 2014.

Choice A is incorrect. This is the population of Iceland in 2014. Choice B is incorrect and may result from dividing 3.3 by 2.5, instead of subtracting 2.5 from 3.3. Choice C is incorrect and may result from dividing the population of Iceland in 1990 by 2.

▼ 67 — B

Choice B is correct. It's given that at a particular track meet, the ratio of coaches to athletes is 1 to 26. If one number in a ratio is multiplied by a value, the other number must be multiplied by the same value in order to maintain the same ratio. If there are x coaches at the track meet, multiplying both numbers in the ratio by x yields $1x$ to $26x$, or x to $26x$. Therefore, the expression $26x$ represents the number of athletes at the track meet.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 68 — 34672

The correct answer is $34{,}672$. It's given that $1 \text{ mile} = 1{,}760 \text{ yards}$. It follows that an average speed of 19.700 miles per hour is equivalent to $\frac{19.700 \text{ miles}}{1 \text{ hour}} \frac{1{,}760 \text{ yards}}{1 \text{ mile}}$, or $34{,}672$ yards per hour.

▼ 69 — D

Choice D is correct. Since 1 minute = 60 seconds and 1 hour = 60 minutes, it follows that 1 hour = (60)(60), or 3,600 seconds. Using this conversion factor, the space station's average speed of 4.76 miles per second is equal to an average speed of $\frac{4.76 \text{ miles}}{\text{second}} \times \frac{3{,}600 \text{ seconds}}{\text{hour}} = \frac{17{,}136 \text{ miles}}{\text{hour}}$, or 17,136 miles per hour.

Choice A is incorrect. This is the space station's average speed in miles per minute. Choice B is incorrect. This is double the space station's average speed in miles per minute, or the number of miles the space station travels on average in 2 minutes. Choice C is incorrect. This is triple the space station's average speed in miles per minute, or the number of miles the space station travels on average in 3 minutes.

▼ 70 — 1015

The correct answer is 1,015. It's given that the first moon orbits the planet every 252 days. Therefore, it takes the first moon 25229, or 7,308, days to orbit the planet 29 times. It's also given that the second moon orbits the planet every 287 days. Therefore, it takes the second moon 28729, or 8,323, days to orbit the planet 29 times. Since it takes the first moon 7,308 days and the second moon 8,323 days, it takes the second moon 8,323 - 7,308, or 1,015, more days than it takes the first moon to orbit the planet 29 times.

▼ 71 — 14/3, 4.666, 4.667

The correct answer is $\frac{14}{3}$. It's given that 3 teaspoons is equivalent to 1 tablespoon. Therefore, 14 teaspoons is equivalent to $14 \text{ teaspoons} \frac{1 \text{ tablespoon}}{3 \text{ teaspoons}}$, or $\frac{14}{3}$ tablespoons. Note that 14/3, 4.666, and 4.667 are examples of ways to enter a correct answer.

▼ 72 — 40260

The correct answer is 40,260. It's given that $1 \text{ furlong} = 220 \text{ yards}$ and $1 \text{ yard} = 3 \text{ feet}$. It follows that a distance of 61 furlongs is equivalent to $61 \text{ furlongs} \frac{220 \text{ yards}}{1 \text{ furlong}} \frac{3 \text{ feet}}{1 \text{ yard}}$, or 40,260 feet.

▼ 73 — A

Choice A is correct. Since the distance is equal to the amount of time multiplied by a constant, the given equation $d = 55t$ represents a proportional relationship between distance and time in this situation. Since $9k = 3 \cdot 3k$, the time when $t = 9k$ hours is 3 times the time when $t = 3k$ hours. Therefore, the distance traveled after $9k$ hours is 3 times the distance after $3k$ hours.

Choices B and D are incorrect and may result from interpreting the proportional relationship between time and distance as additive rather than multiplicative. Choice C is incorrect and may result from an arithmetic error.

▼ 74 — ?

The correct answer is 25.4. The average speed is the total distance divided by the total time. The total distance is 11 miles and the total time is 26 minutes. Thus, the average speed is $\frac{11}{26}$ miles per minute. The question asks for the average speed in miles per hour,

and there are 60 minutes in an hour; converting miles per minute to miles per hour gives the following:

$$\text{Average speed} = \frac{11 \text{ miles}}{26 \text{ minutes}} \times \frac{60 \text{ minutes}}{1 \text{ hour}}$$

$$= \frac{660}{26} \text{ miles per hour}$$

$$\approx 25.38 \text{ miles per hour}$$

Therefore, to the nearest tenth of a mile per hour, the average speed of Paul Revere's ride would have been 25.4 miles per hour. Note that 25.4 and 127/5 are examples of ways to enter a correct answer.

▼ 75 — A

Choice A is correct. It's given that $1 \text{ foot} = 12 \text{ inches}$. It follows that $1^2 \text{ square foot} = 12^2 \text{ square inches}$, or $1 \text{ square foot} = 144 \text{ square inches}$. Thus, $3{,}168$ square inches is equivalent to $(3{,}168 \text{ square inches})\frac{1 \text{ square foot}}{144 \text{ square inches}}$, or 22 square feet.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the value of $\frac{3{,}168}{12}$, not $\frac{3{,}168}{144}$.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 76 — D

Choice D is correct. It's given that on average, a certain plant grows 87 millimeters every m months. This average rate can be written as $\frac{87}{m}$ millimeters per month. Since there are 12 months in 1 year, it follows that there are $12k$ months in k years. Therefore, the number of millimeters, on average, the plant grows every k years can be calculated by multiplying the rate $\frac{87}{m}$ millimeters per month by $12k$ months, which yields $\frac{87}{m}(12k)$, or $\frac{1{,}044k}{m}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 77 — A

Choice A is correct. It's given that 1 meter $= 3.28$ feet. It follows that 1^2 square meter $= 3.28^2$ square feet, or 1 square meter $= 10.7584$ square feet. Since 1 hour $= 60$ minutes, it follows that 250 square feet per hour is equivalent to $\frac{250 \text{ square feet}}{1 \text{ hour}} \frac{1 \text{ square meter}}{10.7584 \text{ square feet}} \frac{1 \text{ hour}}{60 \text{ minutes}}$, or $\frac{250 \text{ square meters}}{645.504 \text{ minutes}}$, which is approximately 0.3873 square meters per minute. Of the given choices, 0.39 is closest to 0.3873.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 78 — B

Choice B is correct. It's given that the density of a certain type of wood is 353 kilograms per cubic meter kg/m^3, and a sample of this type of wood has a mass of 345 kg. Let x represent the volume, in m^3, of the sample. It follows that the relationship between the density, mass, and volume of this sample can be written as $\frac{353 \text{ kg}}{1 \text{ m}^3} = \frac{345 \text{ kg}}{x \text{ m}^3}$, or $353 = \frac{345}{x}$. Multiplying both sides of this equation by x yields $353x = 345$. Dividing both sides of this equation by 353 yields $x = \frac{345}{353}$. Therefore, the volume of this sample is $\frac{345}{353}$ m^3. Since it's given that the sample of this type of wood is a cube, it follows that the length of one edge of this sample can be found using the volume formula for a cube, $V = s^3$, where V represents the volume, in m^3, and s represents the length, in m, of one edge of the cube. Substituting $\frac{345}{353}$ for V in this formula yields $\frac{345}{353} = s^3$. Taking the cube root of both sides of this equation yields $\sqrt[3]{\frac{345}{353}} = s$, or $s \approx 0.99$. Therefore, the length of one edge of this sample to the nearest hundredth of a meter is 0.99.

Choices A, C, and D are incorrect and may result from conceptual or calculation errors.

▼ 79 — A

Choice A is correct. It's given that the sample is in the shape of a cube with edge lengths of 0.60 meters. Therefore, the volume of the sample is 0.60^3, or 0.216, cubic meters. It's also given that the sample has a density of 813 kilograms per 1 cubic meter. Therefore, the mass of this sample is 0.216 cubic meters $\frac{813 \text{ kilograms}}{1 \text{ cubic meter}}$, or 175.608 kilograms. Rounding this mass to the nearest whole number gives 176 kilograms. Therefore, to the nearest whole number, the mass, in kilograms, of this sample is 176.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 80 — D

Choice D is correct. It's given that Anita used a ratio of 2 ounces of blue paint to 3 ounces of yellow paint for the first batch. For any batch of paint that uses the same ratio, the amount of yellow paint used will be $\frac{3}{2}$, or 1.5, times the amount of blue paint used in the batch. Therefore, the amount of yellow paint Anita will use in the second batch will be 1.5 times the amount of blue paint used in the second batch.

Alternate approach: It's given that Anita used a ratio of 2 ounces of blue paint to 3 ounces of yellow paint for the first batch and that she will use 5 ounces of blue paint for the second batch. A proportion can be set up to solve for x, the amount of yellow paint she will use for the second batch: $\frac{2}{3} = \frac{5}{x}$. Multiplying both sides of this equation by 3 yields $2 = \frac{15}{x}$, and multiplying both sides of this equation by x yields $2x = 15$. Dividing both sides of this equation by 2 yields $x = 7.5$. Since Anita will use 7.5 ounces of yellow paint for the second batch, this is $\frac{7.5}{5} = 1.5$ times the amount of blue paint (5 ounces) used in the second batch.

Choices A, B, and C are incorrect and may result from incorrectly interpreting the ratio of blue paint to yellow paint used.

▼ 81 — D

Choice D is correct. It's given that a hose puts $88x$ ounces of water in a bucket in $5y$ minutes. Therefore, the rate at which the hose puts water in the bucket, in ounces per minute, can be represented by the expression $\frac{88x}{5y}$. Let w represent the number of ounces of water the hose puts in the bucket in $9y$ minutes at this rate. It follows that the rate at which the hose puts water in the bucket, in ounces per minute, can be represented by the expression $\frac{w}{9y}$. The expressions $\frac{88x}{5y}$ and $\frac{w}{9y}$ represent the same rate, so it follows that $\frac{88x}{5y} = \frac{w}{9y}$. Multiplying both sides of this equation by $9y$ yields $\frac{792xy}{5y} = w$, or $\frac{792x}{5} = w$. Therefore, the number of ounces of water the hose puts in the bucket in $9y$ minutes can be represented by the expression $\frac{792x}{5}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 82 — B

Choice B is correct. It's given that object R travels a distance of $4x$ inches in y seconds. This speed can be written as $\frac{4x \text{ inches}}{y \text{ seconds}}$. It's given that the speed of object R is half the speed of object S. It follows that the speed of object S is twice the speed of object R, which is $2\frac{4x \text{ inches}}{y \text{ seconds}}$, or $\frac{8x \text{ inches}}{y \text{ seconds}}$. Let n represent the time, in seconds, it takes object S to travel a distance of $24x$ inches. The value of n can be found by solving the equation $\frac{8x \text{ inches}}{y \text{ seconds}} = \frac{24x \text{ inches}}{n \text{ seconds}}$, which can be written as $\frac{8x}{y} = \frac{24x}{n}$. Multiplying each side of this equation by ny yields $8xn = 24xy$. Dividing each side of this equation by $8x$ yields $n = 3y$. Therefore, the expression $3y$ represents the time, in seconds, it takes object S to travel a distance of $24x$ inches.

Choice A is incorrect. This expression represents the time, in seconds, it would take object S to travel a distance of $24x$ inches if the speed of object R were twice, not half, the speed of object S.

Choice C is incorrect. This expression represents the time, in seconds, it takes object S to travel a distance of $128x$ inches, not $24x$ inches.

Choice D is incorrect. This expression represents the time, in seconds, it takes object R, not object S, to travel a distance of $24x$ inches.

▼ 83 — 0.32, 8/25

The correct answer is $.32$. The volume of a cube is given by the formula $V = s^3$, where s is the length of an edge of the cube. It's given that each edge of the cube has a length of 2.00 centimeters. Substituting 2.00 centimeters for s in the formula $V = s^3$ yields $V = (2.00 \text{ centimeters})^3$, or $V = 8.00$ cubic centimeters. It's given that the cube has a mass of 2.56 grams. Dividing the mass, in grams, of the cube by the volume, in cubic centimeters, of the cube gives its density, in grams per cubic centimeters. Therefore, the density of the cube is $\frac{2.56 \text{ grams}}{8.00 \text{ cubic centimeters}}$, or $.32$ grams per cubic centimeter. Note that .32 and 8/25 are examples of ways to enter a correct answer.

▼ 84 — A

Choice A is correct. It's given that the sample is in the shape of a cube with edge lengths of 0.9 meters. Therefore, the volume of the sample is 0.90^3, or 0.729, cubic meters. It's also given that the sample has a density of 807 kilograms per 1 cubic meter. Therefore, the mass of this sample is $0.729 \text{ cubic meters} \frac{807 \text{ kilograms}}{1 \text{ cubic meter}}$, or 588.303 kilograms. Rounding this mass to the nearest whole number gives 588 kilograms. Therefore, to the nearest whole number, the mass, in kilograms, of this sample is 588.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 85 — ?

The correct answer is 30. The situation can be represented by the equation $x(2^4) = 480$, where the 2 represents the fact that the amount of money in the account doubled each year and the 4 represents the fact that there are 4 years between January 1, 2001, and January 1, 2005. Simplifying $x(2^4) = 480$ gives $16x = 480$. Therefore, $x = 30$.

▼ 86 — 6.21

The correct answer is 6.21. It's given that the samples of pumice were cut in the shape of a cube. It's also given that the length of the edge of one of these cubes is 3.000 centimeters. Therefore, the volume of this cube is $3.000 \text{ centimeters}^3$, or 27 cubic centimeters. Since the density of this cube is 0.230 grams per cubic centimeter, it follows that the mass of this cube is $\frac{0.230 \text{ grams}}{1 \text{ cubic centimeter}} 27 \text{ cubic centimeters}$, or 6.21 grams.

▼ 87 — .0625, 1/16

The correct answer is .0625 . It's given that $\frac{4a}{b} = 6.7$ and $\frac{a}{bn} = 26.8$. The equation $\frac{4a}{b} = 6.7$ can be rewritten as $4\frac{a}{b} = 6.7$. Dividing both sides of this equation by 4 yields $\frac{a}{b} = 1.675$. The equation $\frac{a}{bn} = 26.8$ can be rewritten as $\frac{a}{b}\frac{1}{n} = 26.8$. Substituting 1.675 for $\frac{a}{b}$ in this equation yields $1.675\frac{1}{n} = 26.8$, or $\frac{1.675}{n} = 26.8$. Multiplying both sides of this equation by n yields $1.675 = 26.8n$. Dividing both sides of this equation by 26.8 yields $n = 0.0625$. Therefore, the value of n is 0.0625. Note that .0625, 0.062, 0.063, and 1/16 are examples of ways to enter a correct answer.

▼ 88 — C

Choice C is correct. It's given that in a scale model, a length of 1 inch is equivalent to a length of $\frac{17}{3}$ feet in the actual office building. This means that the ratio of a length, in feet, in the actual office building to the corresponding length, in inches, in the model is $\frac{17}{3}$ to 1. Therefore, the length, in inches, in the model corresponding to a length in the actual office building of x feet can be found by dividing x by $\frac{17}{3}$. Dividing by $\frac{17}{3}$ is equivalent to multiplying by $\frac{3}{17}$, so the corresponding length, in inches, in the model in terms of x can be represented by the expression $\frac{3x}{17}$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect. This expression represents the corresponding length, in inches, in the model in terms of x for a model where 1 inch in the model is equivalent to a length of $\frac{3}{17}$, not $\frac{17}{3}$, feet in the actual building.

▼ 89 — 4176

The correct answer is $4{,}176$. It's given that the side length of the larger square is 3 times the side length of the smaller square. This means that the area of the larger square is 3^2, or 9, times the area of the smaller square. If the area of the smaller square is represented by x, then the area of the larger square can be represented by $9x$. Therefore, the flat surface of the two adjacent squares has a total area of $x + 9x$, or $10x$. It's given that an electric field with strength 29.00 volts per meter passes uniformly through this surface and the total electric flux of the electric field through this surface is $4{,}640 \text{ volts} \cdot \text{meters}$. Since it's given that the electric flux is the product of the electric field's strength and the area of the surface, the equation $29.0010x = 4{,}640$, or $290x = 4{,}640$, can be used to represent this situation. Dividing each side of this equation by 290 yields $x = 16$. Substituting 16 for x in the expression for the area of the larger square, $9x$, yields 916, or 144, square meters. Since the area of the larger square is 144 square meters, the electric flux, in $\text{volts} \cdot \text{meters}$, of the electric field through the larger square can be determined by multiplying the area of the larger square by the strength of the electric field. Thus, the electric flux is $144 \text{ square meters} \frac{29.00 \text{ volts}}{\text{meter}}$, or $4{,}176 \text{ volts} \cdot \text{meters}$.

▼ 90 — D

Choice D is correct. Since the number of yards in 1 mile is $1{,}760$, the number of square yards in 1 square mile is $(1{,}760)(1{,}760) = 3{,}097{,}600$. Therefore, if the area of the town is 4.36 square miles, it is $4.36(3{,}097{,}600) = 13{,}505{,}536$, in square yards.

Choice A is incorrect and may result from dividing the number of yards in a mile by the square mileage of the town.

Choice B is incorrect and may result from multiplying the number of yards in a mile by the square mileage of the town.

Choice C is incorrect and may result from dividing the number of square yards in a square mile by the square mileage of the town.

▼ 91 — 157.8, 789/5

The correct answer is 157.8. It's given that an area of 46.00 square nautical miles is equivalent to k square kilometers. Since 1 nautical mile is equal to 1.852 kilometers, it follows that an area of 1 square nautical mile is equivalent to $(1.852)^2$, or 3.429904, square kilometers. Multiplying 46.00 by 3.429904 yields 157.775584 square kilometers. Rounding this value to the nearest tenth yields 157.8. Therefore, the value of k is 157.8.

▼ 92 — B

Choice B is correct. Since 1 mile is equal to $1{,}760$ yards, 1 square mile is equal to $1{,}760^2$, or $3{,}097{,}600$, square yards. It's given that the park has an area of $11{,}863{,}808$ square yards. Therefore, the park has an area of $11{,}863{,}808 \text{ square yards} \frac{1 \text{ square mile}}{3{,}097{,}600 \text{ square yards}}$, or $\frac{11{,}863{,}808}{3{,}097{,}600}$ square miles. Thus, the area, in square miles, of the park is 3.83.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the square root of the area of the park in square yards, not the area of the park in square miles.

Choice D is incorrect and may result from converting $11{,}863{,}808$ yards to miles, rather than converting $11{,}863{,}808$ square yards to square miles.

▼ 93 — B

Choice B is correct. It's given that the speed of a vehicle is increasing at a rate of 7.3 meters per second squared. It's given to use $1 \text{ mile} = 1{,}609 \text{ meters}$. There are 60 seconds in 1 minute; therefore, 60^2 or $3{,}600$ seconds squared is equal to 1 minute squared. It follows that the rate of 7.3 meters per second squared is equivalent to $\frac{7.3 \text{ meters}}{1 \text{ second squared}} \frac{1 \text{ mile}}{1{,}609 \text{ meters}} \frac{3{,}600 \text{ seconds squared}}{1 \text{ minute squared}}$, or approximately $16.33 \text{ miles per minute squared}$. The rate, in miles per minute squared, rounded to the nearest tenth is 16.3.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.